

INTRODUCTION TO COMPUTER SYSTEM, NETWORKS AND SECURITY

Assignment 4 – Cybersecurity

Part 3 - Network Security Report – Production & IoT (192.168.100.0/23)

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Purpose for the whole assignment 4:

- ❖ Students will analyse and discuss a recent vulnerability or cybersecurity attack. This demonstrates an understanding of related cybersecurity topics and the ability to research information on cybersecurity incidents.
- ❖ Students will write a report on how a specified set of security controls is implemented in a medium-sized enterprise scenario. This report shows the understanding of various security controls in the departments and the ability to evaluate and explain their utilisation.
- ❖ Students will examine the security of two or more apps in the same category on their mobile device, such as Entertainment, Shopping, Education, Messaging, Banking, Games, or Social Networks, showing an understanding of mobile security, privacy and ethical concerns.

Purpose for the assignment 4 part 3:

TASK	BRIEF DESCRIPTION
Provide a report demonstrating how a specific set of security controls is utilised within a medium-sized enterprise scenario	In this task, you assume the role of a security architect (as defined in the NIST NICE workforce framework ²). You are responsible for redesigning a company network (using best practices, as outlined in the NSA Network Infrastructure Security Guide), including placing security controls in the appropriate locations within the network. As security always incurs costs, prepare a report that explains to company management why each security control is required at a particular part of the company network.

Network Security Design for Automated Vacuum Cleaner Production – Department: Production & IoT Devices

Overview

This report presents a secure network design for the Production & IoT department (subnet 192.168.100.0/23, ~300 devices). It protects 24/7 automated machines, production-control PCs, and supporting services while enabling scheduled data transfers and limiting administrative access. The design follows industry best practices (segmentation, least privilege, defence-in-depth) and explains why each control is placed where it is.

Entities used by the Production & IoT department (precise identification)

Production-control PCs (industrial Windows/Linux) run machine-control software and receive design files every 6 hours. Production machines, PLCs, and IoT controllers connect via an edge PoE switch to a core switch linked to the datacenter firewall. Key servers include Authentication (RADIUS/AD), DNS, Management (schedules/config/logs), Monitoring (SIEM/IDS), and Backup/Storage. The OT VLAN isolates production traffic. A VPN gateway provides secure admin access to non-production networks.

Security controls: purpose & justification

Network segmentation (VLANs/ACLs) isolates OT from IT networks. Firewalls enforce strict, allowlisted traffic. VPN with MFA secures remote admin access. TLS protects file transfers and web interfaces. Centralized authentication with salted password hashing enforces strong policies. Disk encryption secures portable/admin devices. WPA3-Enterprise with 802.1X protects Wi-Fi. Immutable backups enable ransomware recovery while maintaining high availability.

Network Diagrams

Below are two diagrams provided in Mermaid syntax (paste into a Mermaid-enabled editor or Lucidchart with Mermaid import). They represent: network layout without security controls, and network layout with the security controls implemented.

Diagram without Security control

Planning:

```
Internet((Internet)) → EdgeRouter[Edge Router]
EdgeRouter → CoreSwitch[Core Switch]
CoreSwitch → ProdSwitch[Production/OT Switch]
CoreSwitch → AdminSwitch[Admin Switch]
ProdSwitch → PLCs[Production Machines / PLCs]
ProdSwitch → ProdPCs[Production-control PCs]
AdminSwitch → AdminPCs[Admin PCs & Laptops]
AdminSwitch → AdminServer[Admin Server & DB]
EdgeRouter → WebServer[Web Server]
EdgeRouter → MailServer[Mail Server]
AdminSwitch → WiFiAP[Wi-Fi APs]
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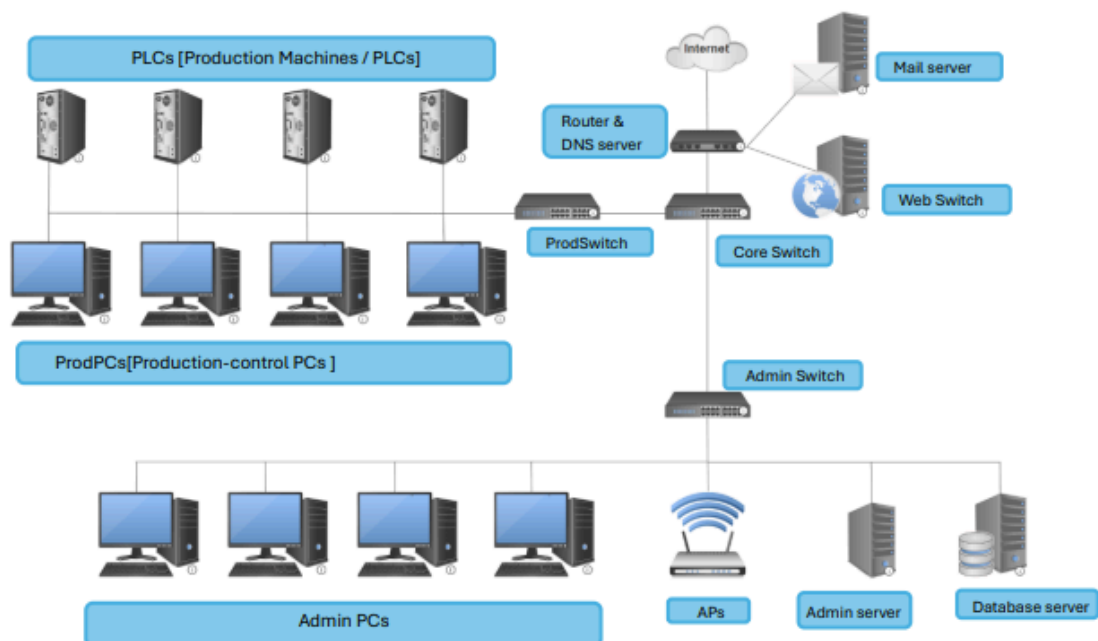


Diagram with security control

Planning:

Internet((Internet))→ EdgeFW[Perimeter Firewall & Edge Router]

EdgeFW → DMZ[DMZ]

DMZ → WebServer[Web Server (HTTPS/TLS)]

DMZ → MailServer[Mail Server (TLS)]

EdgeFW → CoreSwitch[Core Switch]

CoreSwitch → AdminFW[Internal Firewall - IT/OT Segmentation]

CoreSwitch→ Monitoring[SIEM/IDS Collector]

AdminFW → AdminSwitch[Admin Switch (Admin VLANs)]

AdminSwitch → AdminServer[Admin Server / DB (TLS)]

AdminSwitch → AdminPCs[Admin PCs & Laptops (Disk encryption)]

AdminSwitch → WiFiGW[Wi-Fi Controller]

WiFiGW→ WiFiAPs[Wi-Fi APs (WPA3, 802.1X)]

AdminFW → JumpHost[Hardened Jump Host (MFA + VPN)]

JumpHost → ProdFW[OT Firewall / VLAN Gateway]

ProdFW → ProdSwitch[Production/OT Switch (192.168.100.0/23)]

ProdSwitch → ProdPCs[Production-control PCs (SFTP/TLS)]

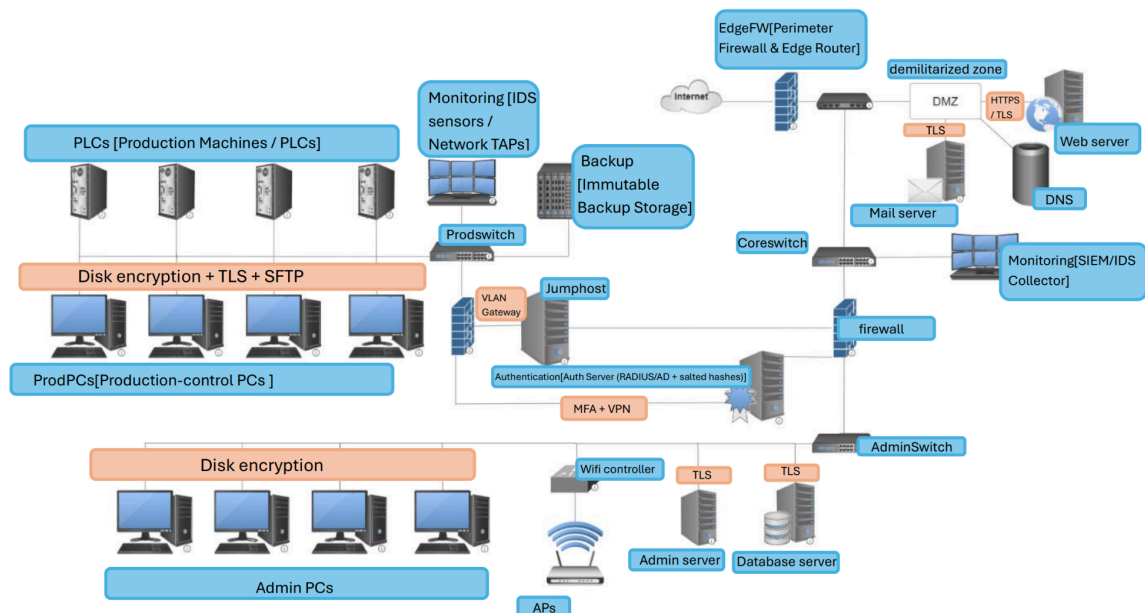
ProdSwitch → PLCs[Production Machines / PLCs]

ProdSwitch → Monitoring[IDS sensors / Network TAPs]

ProdSwitch → Backup[Immutable Backup Storage]

Authentication[Auth Server (RADIUS/AD + salted hashes)] --> AdminFW

Authentication → ProdFW



Notes: The OT firewall only allows specific flows: scheduled file transfer (SFTP/HTTPS) from the management server to production-control PCs on predefined ports and from known IPs, and monitoring/telemetry outbound to the SIEM. Remote admin access must pass through VPN → Jump Host → MFA and is logged.

IP address allocation (192.168.100.0/23)

The /23 gives 512 addresses (192.168.100.0 - 192.168.101.255). We'll allocate to support ~300 devices with room for growth and reserved ranges for infrastructure.

Range	Purpose	Size	Notes
192.168.100.1 - 192.168.100.10	Infrastructure (gateway, DNS, auth, monitoring)	10 IPs	Static. e.g., .1 = ProdFW, .2 = DNS, .3 = Auth server, .4 = SIEM
192.168.100.11 - 192.168.100.50	Servers & printers (static)	40 IPs	Backup, management server, NTP, etc.
192.168.100.51 - 192.168.100.254	Production-control PCs (DHCP reserved pool for static mapping)	204 IPs	Map each control PC by MAC using DHCP reservations
192.168.101.1 - 192.168.101.100	Production machines / PLCs (static)	100 IPs	Many PLCs require static IPs; reserve block for device types
192.168.101.101 - 192.168.101.200	Spare / future expansion	100 IPs	Room for growth, cameras, sensors
192.168.101.201 - 192.168.101.250	Monitoring and management agents	50 IPs	IDS sensors, Taps, Out-of-band mgmt
192.168.101.251 - 192.168.101.254	Broadcast/gateway reserved	4 IPs	Reserved for future infrastructure

Justification: The /23 subnet provides ample addresses for ~300 OT devices with future headroom. Infrastructure devices (gateway, auth, DNS, SIEM) are in the low addresses for consistent routing and easier ACL rules. Servers and printers are statically assigned to simplify certificate issuance and firewall rules. Production-control PCs use DHCP with MAC-to-IP reservations to ease management while preserving predictable addressing for firewall and monitoring. PLCs and embedded devices often prefer static addressing; therefore a dedicated contiguous block (192.168.101.1–100) is reserved and documented, making it simpler to apply ACLs based on IP ranges. Separating servers, PC pool, and PLCs by subranges also simplifies network monitoring, IP-based IDS rules, and incident response. Reserved ranges ensure growth without reworking address plans.

Conclusion

By applying segmentation, strict firewalling, encrypted transports, centralized authentication, and layered monitoring, the Production & IoT network maintains high availability while substantially reducing risk. These controls prioritize uptime for 24/7 operations while enabling required data transfers and safe administrative access.

References

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Appendix

Word count:

Section:	No of word including heading
Title	12
Overview	61
Entities used by the Production & IoT department	140
Network Diagrams	88
IP address allocation	156
Conclusion	41

Therefore the total number of words is $12+61+88+156+41+140 = 498$

Note: cover page, table and planning for diagrams not included